

Hanyu Wang

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EDUCATION

- **Yale University | Master of Science** New Haven, CT, USA
Epidemiology of Infectious Diseases (Quantitative Specialization) *Expected May 2026*
Relevant Coursework: *Quantitative Methods for Infectious Disease (R), Transmission Dynamic Models for Infectious Diseases (R), Advanced Regression Model (R & Python), Multivariate Statistics (R & SAS), Global Change and Infectious Disease*
- **University of Toronto | Honours Bachelor of Science** Toronto, Canada
Statistics Major, Quantitative Biology Major *Jun 2025*
Relevant Coursework: *Time Series Analysis (R), Methods for Multivariate Data (R), Machine Learning (R & Python), Experimental Analysis (R), Population Ecology (R), Computer Programming (Python)*

WORK EXPERIENCE

- **Mosquito-Borne Disease Risk and Climate Vulnerability in Connecticut** *Jan 2026-Present*
Yale Center on Climate Change and Health, Yale University
 - Contributed to the *Climate Change and Health in Connecticut 2025 Report (Infectious Diseases section)*.
 - Synthesized CAES mosquito surveillance reports and CDC epidemiological data to characterize climate-related trends in **vector dynamics and disease risk**.
 - Conducted structured literature and report review to identify environmental drivers of **climate vulnerability** and **regional/population disparities**.
- **Land-Use Change and Tick-Borne Disease Risk** *Nov 2025-Present*
Yale School of the Environment, Yale University
 - Conduct a systematic review and meta-analysis evaluating the impact of land use and land-use change on **tick populations and tick-borne disease risk**.
 - Collaborate with members of Michelle L. Bell's lab to synthesize causal evidence on environmentally mediated transmission pathways.
 - Address critical gaps in quantitative evidence linking ecological change to **vector ecology and disease emergence**.
- **Multifactorial Compartmental Model for Infectious Disease Dynamics** *May-Aug 2024*
Laboratory for Industrial and Applied Mathematics, York University
 - Developed an optimized **SEIRS compartmental model** to simulate infectious disease transmission, incorporating demographic and biological factors.
 - Integrated **stochasticity and spatial dynamics**, explicitly modeling birth, death, and immune decay via computational simulations.
 - Collaborated with a team of four to refine model accuracy; enhanced expertise in **mathematical modeling and public health informatics**.
- **Single-Cell Transcriptome Sequencing Data Analysis** *Sep-Dec 2024*
Biological Diagnosis and Treatment Center, Xibei Hospital
 - Analyzed scRNA-seq data from enteritis patients to study gene expression heterogeneity and identify therapeutic cell populations.
 - Built **R pipelines** using **Seurat**, implementing PCA, Louvain/Leiden clustering, and visualization (t-SNE, UMAP).
 - Identified **10 cell subtypes from 19 clusters**, advancing skills in biomedical data analysis and machine learning.

TECHNICAL & RESEARCH SKILLS

Programming & Computational Tools

R (Advanced): statistical modeling, spatiotemporal analysis, data visualization

Python (Proficient): data processing, machine learning, stochastic simulation, scientific computing

Statistical & Mathematical Modeling

Transmission Modeling: structured compartmental models, mechanistic modeling, stochastic and spatially explicit simulations

Inference & Uncertainty: Bayesian hierarchical modeling, likelihood-based inference, parameter estimation, sensitivity analysis

Spatiotemporal Modeling: time-series analysis, spatial models, GLM/GAM, mixed-effects models

Research Domains & Applications

Infectious Disease Epidemiology: transmission dynamics, population-level inference, climate-sensitive disease systems

Disease Ecology: host-pathogen interactions, environmentally mediated transmission, multi-scale dynamics

Spatial Epidemiology: disease mapping, risk modeling, geospatial analysis of vector-borne diseases